MatchPulse — Architecture Description

Date	2026-11-10
Courses	Software Architectures (CM0639 module 1) and Development Methodologies
Code	v3.0.0 plus the v3.1.0 security and integrity remediation (unreleased)
Reading time	A 3-minute walkthrough is below; the reference tables are for questions

1. What MatchPulse is

MatchPulse is a club-first football ticketing and fan-engagement platform. A club subscribes to a plan, creates events and matches, sells named tickets, checks attendees in, and reads aggregate prediction insights for its own matches. Fans use one email/password account to buy tickets and predict with points across clubs. A platform admin manages clubs, plans and audit. Commerce is simulated.

The hard part is not the screen. It is keeping capacity, points, ticket codes, club isolation and resolution correct when many fans act at the same time.

2. Three-minute walkthrough

> MatchPulse is a course prototype for club ticketing and match-day prediction. The user is a club organizer. The problem is that ticket sales, attendee lists, check-in and post-match sentiment are usually spread across spreadsheets and separate tools. At kickoff, an application-level check fails: tickets can be oversold, points can be charged twice, a winning share can be paid twice, or one club can see another club's data.
>
> I built one Next.js application and one PostgreSQL database. The web UI and the CLI are two clients of the same typed domain contract. The database is the system of record and the authorization boundary: RLS, grants and `SECURITY DEFINER` functions check the caller, the club, the role and the plan.
>
> The structure is a modular monolith. The modules are identity and access, clubs and plans, events and matches, ticketing, prediction market, insights and privacy, and admin and audit. They communicate through SQL calls inside one transaction, not through a network. I did not split into microservices because there is one developer, one database and one deployment, and the domain needs strong consistency, not independent scaling.
>
> The main decisions are: Supabase Auth for email/password identity; one database with RLS and transactional functions; club tenancy with database-enforced entitlements; nominal tickets with a buyer and named attendees; and a result-bound, one-shot prediction market.
>
> The trade-off is deliberate. Pessimistic locks protect balances and capacity but reduce peak throughput. Database-centric rules are hard to bypass but tie the design to PostgreSQL. One deployable keeps operations simple but prevents independent scaling. Those are acceptable for this prototype.
>
> What I do not claim: there is no hosted deployment, no production recovery, no real money, no independent user study and no third-party security audit. The evidence is local and CI-based. The architecture is modular enough to extract a module later if a measured driver appears, but I have not done that.

3. Structure



There is no broker, cache, worker or separate service. A request is a synchronous call from the browser or CLI to the Next.js/domain layer, then to a transactional PostgreSQL function. That is enough for the current domain and avoids distributed consistency problems the product does not need.

4. Modules and boundaries

Module	Responsibility	Main storage
Identity and access	Supabase Auth, profile, platform admin flag, membership and role checks	`profiles`, `club_members`
Clubs and plans	Tenancy, invitations, plans, entitlements, subscription status	`clubs`, `plans`, `club_subscriptions`
Events and matches	Draft/publish, capacity, ticket allocation, kickoff, scores	`tournaments`, `matches`

Module	Responsibility	Main storage
Ticketing	Ticket types, nominal orders, attendee identity, code/QR, transfer, check-in, retention	`ticket_types`, `ticket_orders`, `tickets`
Prediction market	LMSR pricing, entry/exit, cooldown, idempotency, resolution	`markets`, `market_positions`, `market_trades`
Insights and privacy	Club-scoped aggregates, cohort suppression, audited export	`club_prediction_insights`, `audit_events`
Admin and audit	Clubs, plans, subscription status, audit; no impersonation	`platform_admins`, `audit_events`

Boundaries. Client components do not import the server Supabase client. Server Actions validate input and call authorized RPCs; they do not mutate tables directly. Mutation RPCs are `SECURITY DEFINER`, resolve `auth.uid()`, fix `search_path` and grant execute only to `authenticated`. Direct table writes are revoked. The service-role key is a fixture/test/maintenance credential, never an application runtime key.

5. Key decisions

Decision	Why	Record
Next.js + Supabase + PostgreSQL	One developer, two course deadlines; database-enforced rules; local tooling	ADR-001
Authentication, RLS, grants and `SECURITY DEFINER` RPCs	Direct writes would bypass business rules; the UI is not the boundary	ADR-003
Club tenancy and database-enforced plans	Clubs are the paying tenant; a hidden button is not an entitlement check	ADR-007
Nominal ticketing with buyer vs attendee	Clubs need named attendees and event-access identity, not a ticket count	ADR-010
Result-bound, one-shot prediction market	Resolution must match the final score and stay terminal	ADR-011
Separate platform admin	Platform operation is not a club role; no impersonation	ADR-008

6. Where the code is

Element	Location
App Router pages	`src/app/` — fan, club, admin, events, market
Server Actions	`src/app/actions/` — auth, market, tickets, v3
Shared domain contract	`src/lib/domain/` — auth, clubs, events, tickets, predictions, analytics, admin
CLI	`src/cli/` — command tree, profiles, JSON output
Database	`supabase/migrations/` — 29 migrations
Tests	`tests/invariants/`, `tests/integration/`, `tests/agent/`, `tests/e2e/`
Architecture guard	`scripts/check-architecture.ts`
CI	`.github/workflows/ci.yml` — five jobs

7. Agile and evolution

This is an **evolutionary architecture**, not a big design up front. The domain contract is stable, but the implementation moves in small slices with tests and fault injection. The five characteristics are fitness functions. Decisions are recorded in ADRs; a change that alters boundaries or ownership needs a new ADR. If a measured driver appears — independent scaling, a different data owner, a regulatory boundary — a module can be extracted. No such driver exists today.

8. What is not claimed

- No hosted deployment, TLS, DNS, external monitoring or alerting.
- No production backup, retention, RTO/RPO or disaster recovery.
- No real payments, KYC, invoices, refunds or settlement.
- No independent user study, security audit or privacy audit.
- No seat maps, wallet passes, turnstile hardware or FIGC/Questura integration.
- M-INV2 is verified by forced-failure rollback, not observed-red.

9. Detail

- Full architecture reference: architecture README
- Requirements: kata and requirements
- Characteristics: 03-architecture-characteristics.md
- C4 views: 05-c4-diagrams.md

- Code map: 04-code-map.md
- ADRs: adr/
- Invariants and evidence: registry
- AI use: ai-use.md